Seamoon Pandey

AI & Backend Engineer

pandeyseamoon05@gmail.com | 9866142928 | github.com/seamoonpandey | linkedin.com/in/seamoonpandey

Tansen-14, Palpa, Nepal

SUMMARY

AI and backend engineer working on machine learning pipelines, backend services, and full-stack applications. Built an AI-assisted security scanner with a DistilBERT classifier and NestJS backend, and currently build Web3 products at Lamida Inc.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** Node.js, NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend:** React, Next.js, Flutter
- **Security:** XSS detection, context-aware payload generation, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

FLAGSHIP PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner — Final-year Capstone, Team of 3 | Code | Report | Demo

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Architected a polyglot microservice system: a **TypeScript**/**NestJS** core orchestrating three **Python**/**FastAPI** AI services over a **Redis**/**BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritize payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.
- Designed the NestJS core: scan lifecycle, crawler, BullMQ queue processor, audit logging, JWT/API-key auth, WebSocket streaming, and HTML/JSON/PDF report generation, backed by unit and e2e test suites.

SELECTED PROJECTS

AI Artifact Recognition Platform — PEC Hackathon | **YOLOv8**, Flask, React

- Built a Flask backend and React dashboard for real-time, YOLOv8-based object recognition with asynchronous inference and structured monitoring endpoints.

Community Connect — OSM Hackathon | Express.js, Flutter

- Developed a full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.

Garva Diary — BMC InnovateX | Ruby on Rails

- Engineered backend APIs on a Rails MVC architecture with structured validation workflows and mobile integration.

EXPERIENCE

Full-Stack Engineer, Lamida Inc. — Full-time | 23 Feb 2026 – Present

- Build full-stack **Node.js** applications for scalable **Web3** products, including a DAO for the **tUSDT** token.
- **Team Lead, Bitfolio** from September 2026.

Freelance Full-Stack Developer — Contract, Remote

- Delivered production full-stack platforms on an **Express.js + React** stack, handling authentication, REST APIs, and third-party integrations:
  - **Jyotisam:** astrology consultation and booking platform (jyotisam.com).
  - **Mahadaan:** donation and charity platform with secure data handling (mahadaan.com).

Mentor, Call for Enthusiasts Program (i-CES)

- Mentored students on building scalable applications with clean backend architecture and development practices.

EDUCATION

Institute of Engineering, Tribhuvan University — 2022 to 2026

Bachelor of Engineering in Computer Engineering, Pashchimanchal (Western Region) Campus, Pokhara

Aggregate: **67.98%**, First Division | Medium of instruction: English